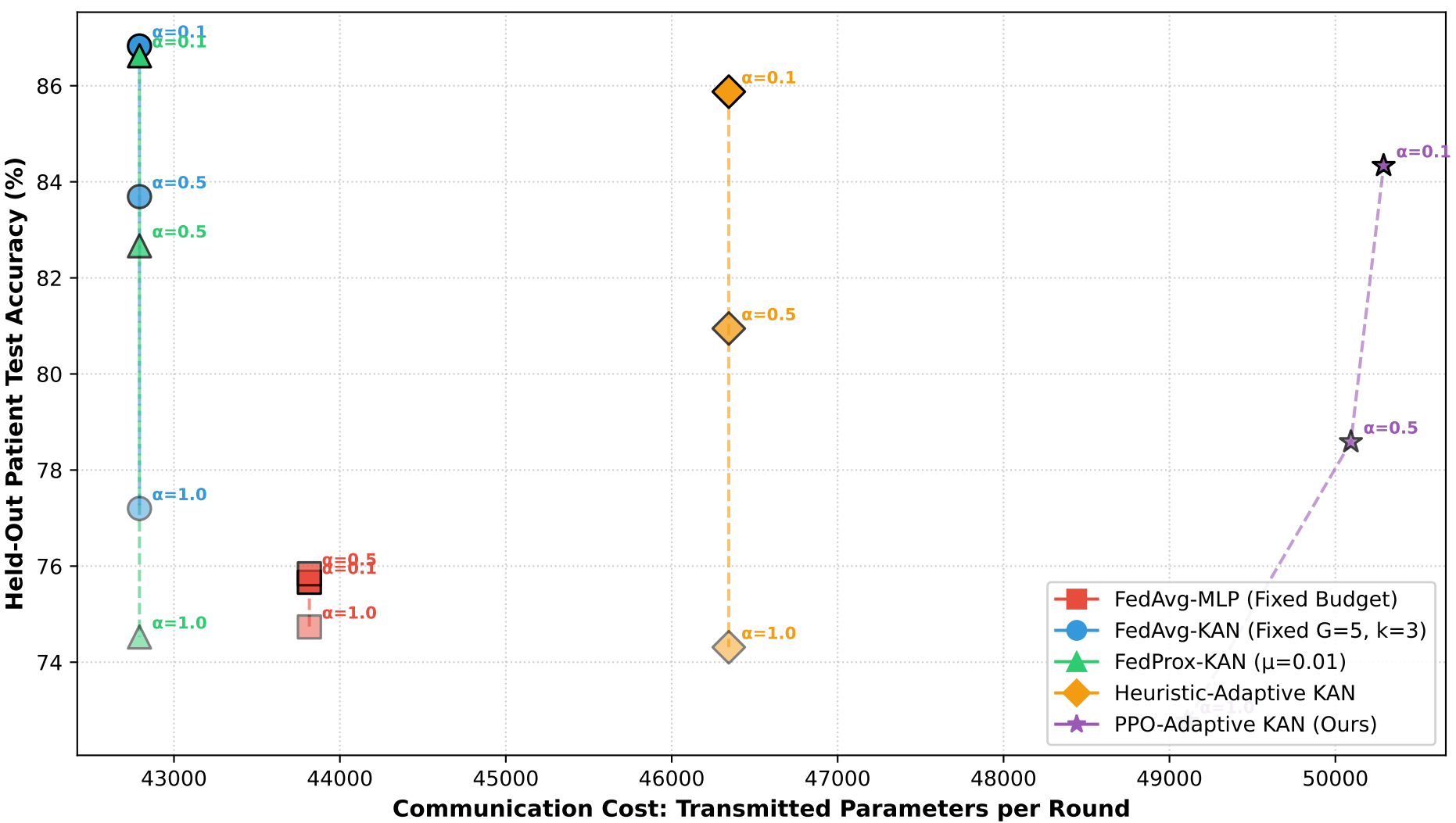
Figure 4.1: Pareto Frontier of Global Accuracy vs. Communication Cost
Held-Out Patient Test Accuracy (%)
Communication Cost: Transmitted Parameters per Round
86
84
82
80
78
76
74
43000
44000
45000
46000
47000
48000
49000
50000
α=0.1
α=0.1
α=0.5
α=0.5
α=1.0
α=1.0
α=0.5
α=0.1
α=1.0
α=0.1
α=0.5
α=1.0
α=0.1
α=0.5
FedAvg-MLP (Fixed Budget)
FedAvg-KAN (Fixed G=5, k=3)
FedProx-KAN (μ=0.01)
Heuristic-Adaptive KAN
PPO-Adaptive KAN (Ours)